



THE MAXWELL EXAMINATIONS BOARD

CANDIDATES' MOCK SERIES II

MATHEMATICS PAPER
DURATION: 2 HOURS 30 MINUTES

INDEX NUMBER

RANDOM NO.						PERSONAL NO.		

CANDIDATE'S NAME: _____

CANDIDATE'S SIGNATURE: _____

DISTRICT NO: _____

Read the following instructions carefully.

1. Do not write your school or district name anywhere on this paper.
2. This paper has two sections: A and B. Section A has 20 questions and section B has 12 questions. The paper has 11 printed pages altogether.
3. Answer all questions. All the answers for both sections A and B must be shown in the spaces provided.
4. All working must be done using a blue or black ball point pen or ink. Any work done in pencil will NOT be marked except graphs and diagrams. Calculators are not allowed in the examination room.
5. Unnecessary changes in your work and handwriting that cannot easily be read may lead to loss of marks.
6. Do not fill anything in the table indicated: "For examiners' use only" and the boxes inside the question paper.

FOR EXAMINERS' USE ONLY

QN NO.	SCORES	EXR'S ID
1-5		
6-10		
11-15		
16-20		
21-22		
23-24		
25-26		
27-28		
29-30		
31-32		
TOTAL		

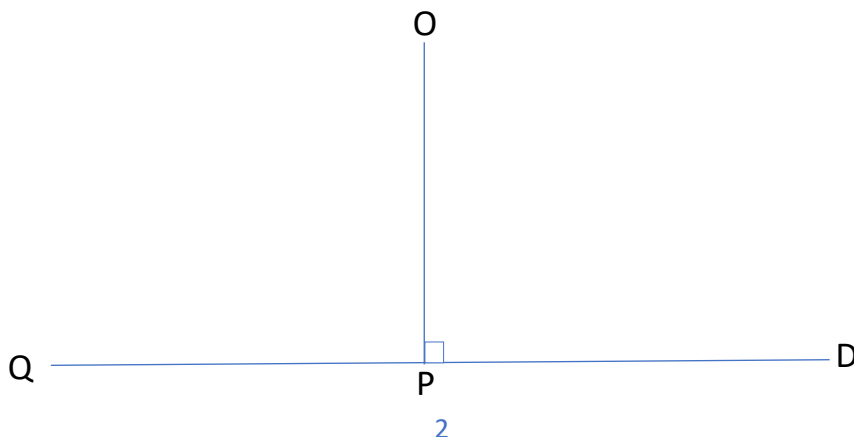
Turn over

SECTION A: 40 MARKS

Questions 1 to 20 carry two marks each.

1. **Work out:** 27×6
2. Find the number represented by:
 $(5 \times 1000) + (3 \times 100) + (8 \times 10) + (4 \times 1)$
3. Given that Set **A** = {2, 4, 6, 8} and Set **B** = {4, 6, 9, 10}, find the number of **A** $\cap$ **B**.
4. A worker earns **sh. 8,000** per day. How much does he earn in a week?
5. Work out the circumference of a circle of radius 7cm. Use **$\pi = 22/7$** .

6. Using a ruler and a pair of compasses only, **bisect** the angle **OPQ** below.



7. **Simplify:** $5x + 3y - 2x - y$

8. **Work out:** $324_{\text{five}} - 143_{\text{five}}$

9. If  represents **50** mangoes, how many mangoes are represented by:    ?

10. **Convert** 7:45 p.m. to the 24-hour clock system.

11. Find the Greatest Common Factor (**GCF**) of 18, 24 and 36.



12. A school had **200** pupils. The enrolment increased by **13%**. Find the new number of pupils.

13. Write the Roman numeral CDXLV in Arabic numerals.

14. Draw and find the area of a rectangle measuring 6cm by 3cm.

15. Simplify: $\frac{1}{2} + \frac{2}{3} - \frac{1}{6}$



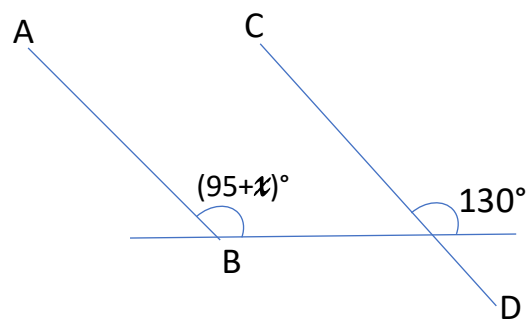
16. Sangora exchanged 250 US dollars into Uganda shillings at sh.3,800 per dollar. How much money did she receive?

17. Find the sum of the missing numbers in the sequence below.

10,, 30, 40,, 60

18. A water tank holds 4,000 litres. How many jerrycans of 20 litres can fill the tank?

19. In the diagram below, line AB is parallel to line CD. Find the value of x .



20. Solve and write the solution set for:

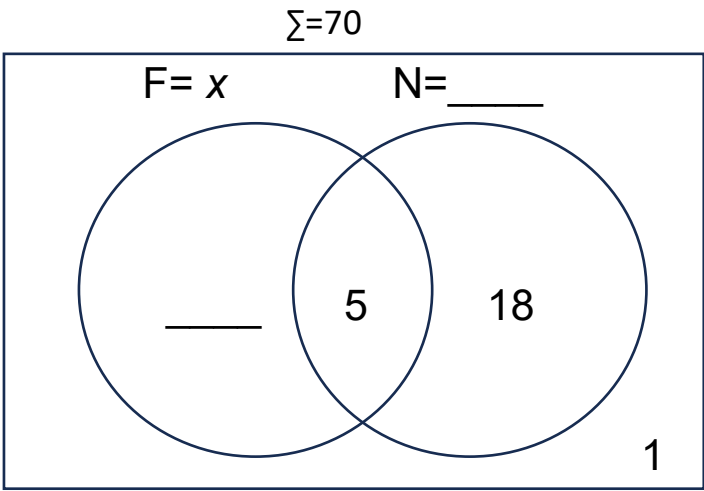
$$m - 3 < 8$$



SECTION B: 60 MARKS

Answer all questions in this section.

21. The Venn diagram below shows pupils who play Football (F) and Netball (N) in P7 Blue at Bright Junior School.



(a) Find the value of x . (02 marks)

(b) Complete the Venn diagram above correctly. (02 marks)

(c) How many pupils participate in a single game? (02 marks)

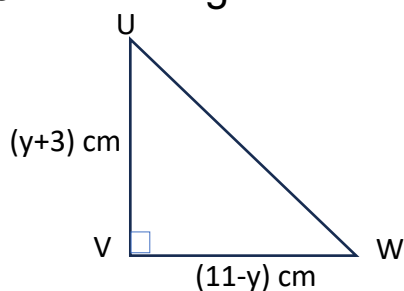
22. Dalton, Whitney and Elizabeth shared oranges in the ratio 2 : 3 : 5 respectively.

(a) Find the total number of oranges if Whitney got 15 oranges. (03 marks)

(b) How many more oranges did Elizabeth get than Dalton? (02 marks)



23. The diagram below shows an isosceles triangle U VW.



(a) Find the value of y . (02 marks)

(b) Find the size of angle U W V. (02 marks)

24. The prime factors of numbers P and Q are:

$$\text{PfP} = \{2^1, 3^1, 5^1\}$$

$$\text{PfQ} = \{2^2, 3^1, 7^1\}$$

(a) Find the Least Common Multiple (LCM) of P and Q. (02 marks)

(b) Work out the value of P and Q. (02 marks)



25. A car travelled at 80km/hr for 3 hours.

(a) Calculate the distance covered. (02 marks)

(b) Another car covered the same distance in 4 hours. Calculate its speed. (03 marks)

26. A woman bought the following items:

2½kg of rice at **sh.6,500**

3 loaves of bread at **sh.5,000 each**

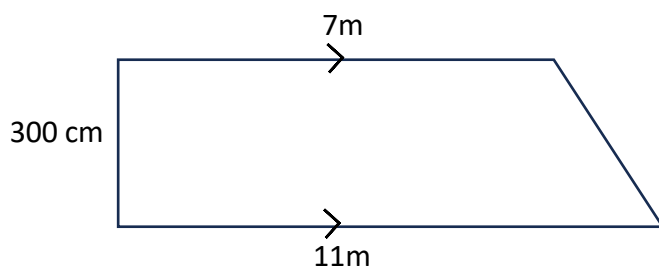
5 litres of milk at **sh.2,500 per litre**

(a) Find her total expenditure. (04 marks)

(b) How much would she spend on 5 loaves of bread? (02 marks)



27. The figure below is a trapezium.



(a) Calculate the area of the trapezium (in metres). (02 marks)

(b) Find the length of the slant side (in metres) using Pythagoras theorem. (03 marks)

28. A bank charges simple interest at 12% per annum.
If sh.900,000 is borrowed for 2 years, calculate:

(a) The interest. (03 marks)

(b) the total amount paid back.

(02 marks)



29. (a) Solve the equation:

$$3w - 4 = 20$$

(03 marks)

(b) Work out:

$$2 + 3 = \underline{\hspace{2cm}} \text{ (finite 5)}$$

(02 marks)

30. The table below shows marks scored by pupils in a test.

Marks	Frequency
40	2
50	3
60	4
70	1

(a) Find the median mark.

(02 marks)

(b) Calculate the average mark.

(03 marks)



31. Using a ruler and a pair of compasses only, construct triangle XYZ in which:

$XY = 7\text{cm}$

$YZ = 5\text{cm}$

$\text{Angle } XYZ = 60^\circ$

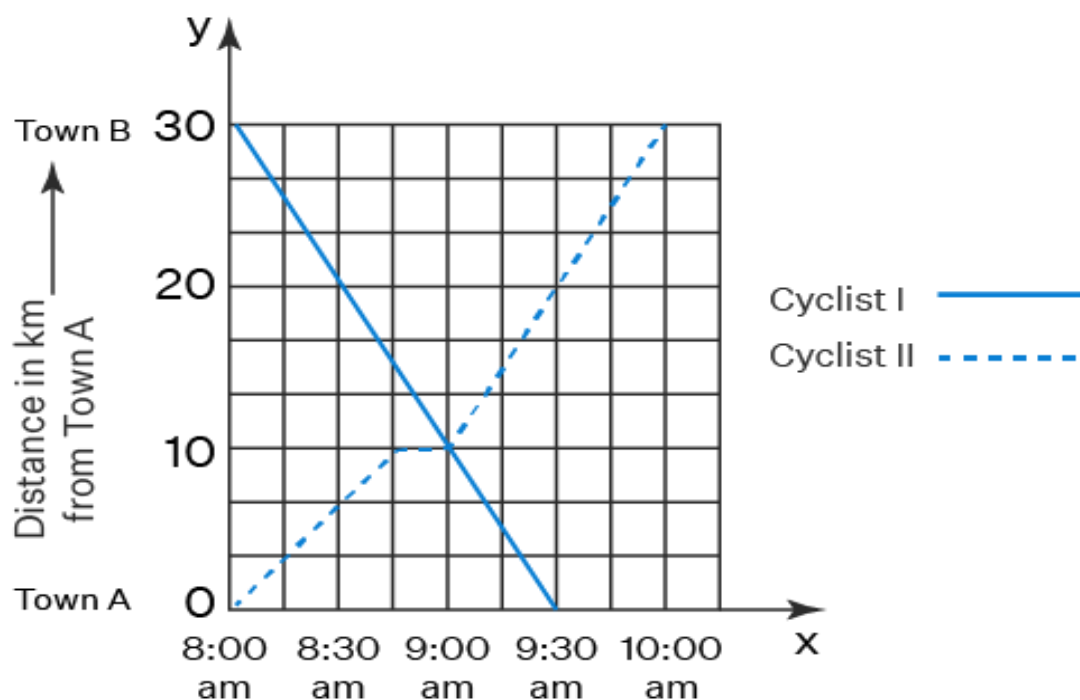
(a) Construct the triangle.

(04 marks)

(b) Measure angle XZY.

(01 mark)

32. Study the distance-time graph below showing **two** journeys. **Cyclist I** travelled from **Town B** to **Town A**, and **Cyclist II** travelled from **Town A** to **Town B**.



- (a) How far is Town B from Town A? (01 mark)
- (b) For how long did **Cyclist II** rest? (02 marks)
- (c) Calculate **Cyclist I's** average speed for the whole journey. (02 marks)

END

